

Draft Paper: Vacuum Response Framework

1. Introduction

Galaxies rotate too fast. Their outer stars move at velocities that cannot be explained by Newtonian gravity sourced by visible matter alone. This empirical fact appears in three closely related forms: the flattening of rotation curves, the baryonic Tully–Fisher relation (BTFR), and the radial acceleration relation (RAR). Each points to a characteristic acceleration scale:

$$a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2.$$

Standard explanations invoke either dark matter or modified gravity, yet neither explains the universality of this scale nor its proximity to cH_0 . This motivates a third possibility: the vacuum contributes to the gravitational source term through a nonlocal response.

2. Motivation

Vacuum energy contributes to curvature in general relativity. If the vacuum responds to global gravitational environments, galaxies may imprint structure onto it, producing an effective gravitational field.

3. Methodological Note

This framework emerged from iterative human–LLM co-analysis, combining conceptual reasoning with rapid structural exploration.

4. The Kernel Theorem

A nonlocal operator acting on baryonic density:

$$\rho_{\text{eff}}(r) = \int K(|r-r'|) \rho(r') d^3r'$$

If $K(k) \propto 1/k$, then $\rho_{\text{eff}}(r) \propto 1/r^2$, yielding flat rotation curves.

5. Nonlinear Normalization

A unique normalization based on global potential depth Ψ leads to:

$$v^4 = G M a_0$$

6. Retarded Dynamics

Causality introduces memory:

$$\blacksquare \Psi = F[\rho]$$

Predicting suppressed response in disturbed galaxies.

7. Cosmological Behavior

The framework admits a de Sitter fixed point:

$$\Psi_\infty = (2/3)c^2$$

8. Kernel Origin

The kernel corresponds to the retarded Green's function of a conformally coupled scalar field in de Sitter space.

9. The Sitter Number

$$S = a_0 / (c H \Lambda)$$

10. Observational Predictions

Redshift, environment, and assembly-history dependence form a distinctive observational signature.

Conclusion

The framework reduces to a single parameter, the Sitter number, and makes testable predictions for current observations.